6-MONTH COMPLETE PROGRAM

DATA ANALYTICS

Python · NumPy · Pandas · SQL · Statistics · Power BI · Tableau · Excel · Gen AI — a complete, structured journey from beginner to job-ready Data Analyst.

6 Months	8 Modules	5+ Projects	Live Sessions	100% Support
Full Duration	Curriculum	Hands-On	Expert-Led	Placement

YOUR LEARNING JOURNEY

A logical, week-by-week path — from Python basics to real dashboards & job-ready projects.

W1–2	W3–4	W5–6	W7	W8–9	W10–11	W12	W13–24
Python Foundations	NumPy & Pandas	SQL	Statistics & Maths	Data Viz	Power BI & Tableau	Excel & Gen AI	Projects & Placement
Variables, OOP, Files	Data Manipulation	Queries & Databases	Probability & Testing	Matplotlib & Seaborn	BI Dashboards	AI Tools & Cloud	Portfolio & Career

COMPLETE CURRICULUM BREAKDOWN

Every topic covered — structured, industry-aligned, and project-driven.

MOD 01 — WEEK 1–2

Python Programming Foundations

- Python Setup: Anaconda, Jupyter, VS Code
- Variables, Data Types & Type Casting
- Operators: Arithmetic, Logical, Comparison
- Control Flow: if/elif/else, for, while
- Functions: Parameters, Return, *args/**kwargs
- Data Structures: List, Tuple, Set, Dict
- OOP: Classes, Objects, Inheritance, Polymorphism
- File Handling: CSV, JSON, TXT
- Error Handling: try / except / finally
- List Comprehensions, Lambda, map/filter

OUTCOME ***Write clean Python programs and solve real data problems from scratch***

MOD 02 — WEEK 3–4

NumPy & Pandas — Data Manipulation

- NumPy Arrays: Creation, Indexing, Slicing
- Reshaping, Broadcasting & Math Operations
- Pandas Series & DataFrame Basics
- Importing CSV, Excel, JSON & SQL Data
- Data Cleaning: Nulls, Duplicates, Outliers
- String Operations & Regex in Pandas
- GroupBy, Aggregations & Pivot Tables
- Merging, Joining & Concatenating DataFrames
- Date & Time Operations
- apply(), map(), applymap()

OUTCOME ***Clean any messy real-world dataset and extract meaningful insights***

MOD 03 — WEEK 5–6

SQL for Data Analysts — Basics to Advanced

- Relational Databases: Tables, Keys, Relations
- DDL: CREATE, ALTER, DROP, TRUNCATE
- DML: SELECT, INSERT, UPDATE, DELETE
- Filtering: WHERE, IN, BETWEEN, LIKE
- Aggregates: COUNT, SUM, AVG + GROUP BY / HAVING
- Joins: INNER, LEFT, RIGHT, FULL, SELF
- Subqueries, Nested Queries & CTEs
- Window Functions: ROW_NUMBER, RANK, LEAD, LAG
- Views, Indexes & Query Optimization
- Stored Procedures, Triggers & Transactions

OUTCOME ***Extract and transform data from any relational database confidently***

MOD 04 — WEEK 7

Statistics & Probability

- Types of Data: Nominal, Ordinal, Continuous
- Central Tendency: Mean, Median, Mode
- Dispersion: Variance, Std Dev, IQR
- Probability: Events, Sample Space, Rules
- Distributions: Normal, Binomial, Poisson
- Central Limit Theorem & Sampling
- Hypothesis Testing: H0 vs H1
- p-value, Confidence Intervals & Alpha
- t-test, Chi-Square Test, ANOVA
- Correlation (Pearson, Spearman) vs Causation

OUTCOME ***Apply statistical thinking to validate findings and present data-backed conclusions***

MOD 05 — WEEK 8–9

Data Visualization — Matplotlib & Seaborn

- Matplotlib: Figure, Axes, Subplots
- Line, Bar, Scatter, Histogram, Pie Charts
- Colours, Labels, Legends, Annotations
- Seaborn: Statistical Plots & Themes
- Distribution Plots: Histplot, KDE, Boxplot
- Relationship Plots: Scatter, Regplot, Pairplot
- Categorical: Barplot, Countplot, Swarmplot
- Heatmaps & Correlation Matrices
- Multi-chart Dashboards with Subplots
- Data Storytelling: Right Chart for Right Data

OUTCOME ***Design clear, impactful visualizations that communicate insights at a glance***

MOD
06
WEEK 10–11

Power BI & Tableau — Business Intelligence

- Power BI Interface & Data Sources
- Power Query: Transformation & Shaping
- Data Modelling: Star & Snowflake Schema
- DAX Fundamentals: Columns, Measures, Tables
- DAX: CALCULATE, FILTER, SUMX, RANKX, DATEADD
- Reports: Slicers, Drill-throughs, Bookmarks
- KPI Cards, Maps, Gantt & Custom Visuals
- Tableau: Sheets, Dashboards & Stories
- Tableau Charts: Bar, Treemap, Bullet Charts
- Publishing to Power BI Service & Tableau Public

OUTCOME ***Build professional, interactive dashboards that executives and managers rely on***

MOD
07
WEEK 12

Excel for Analysts & Gen AI Tools

- VLOOKUP, HLOOKUP, INDEX-MATCH, XLOOKUP
- Pivot Tables, Pivot Charts & Slicers
- Data Validation & Conditional Formatting
- Dynamic Arrays: FILTER, SORT, UNIQUE
- Power Query in Excel: Get & Transform
- Intro to Generative AI for Data Analysts
- ChatGPT Prompting for Analysis & SQL writing
- Copilot in Excel & Power BI AI Visuals
- Using AI for EDA, Reports & Presentations
- AWS Basics & Cloud Data Tools / Snowflake

OUTCOME ***Use Excel and AI tools to become 3x more productive as a Data Analyst***

MOD
08
WEEK 13–24

Real-World Projects, EDA & Career Prep

- Exploratory Data Analysis on 5+ Industry Datasets
- Project 1: Sales Performance Dashboard (Power BI)
- Project 2: Customer Churn Analysis (Python + SQL)
- Project 3: HR Analytics Report (Excel + Tableau)
- Project 4: E-Commerce Analysis (Pandas + Seaborn)
- Capstone: End-to-End Data Analysis Project
- ATS-Friendly Data Analyst Resume Building
- LinkedIn Profile Optimisation for Recruiters
- Mock Interviews: SQL, Python, Stats, Case Studies
- Placement Drive & Company Connect Sessions

OUTCOME ***Graduate with a strong portfolio, polished resume & interview confidence***

TOOLS YOU WILL MASTER

15 industry-standard tools covered across the full program.

Python 3.x	NumPy	Pandas	Matplotlib	Seaborn
Language	Numerics	DataFrames	Charting	Stat Plots

SQL / MySQL	**Power BI**	**Tableau**	**MS Excel**	**Plotly**
Databases	Dashboards	Stories	Spreadsheets	Interactivity
Jupyter	**VS Code**	**GitHub**	**ChatGPT / AI**	**AWS Basics**
Notebooks	IDE	Version Control	Productivity	Cloud

WHO IS THIS FOR?

- **Fresh Graduates**
  BCA, B.Tech, BSc, MBA, BBA — any discipline
- **Career Switchers**
  Come from any background, no prior coding needed
- **Working Professionals**
  Upskill and move into analytics roles
- **Data Enthusiasts**
  Anyone passionate about numbers & insights
- **Entrepreneurs**
  Use data to make smarter business decisions

WHY CODING SHARKS?

- **Industry Expert Mentors**
  Sessions led by working data analysts & engineers
- **Project-Based Learning**
  5+ real-world projects across all modules
- **Full Placement Support**
  Resume, LinkedIn, mock interviews, campus drives
- **Industry Certificate**
  Recognised completion certificate on graduation

Start Your Data Analytics Career Today!

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